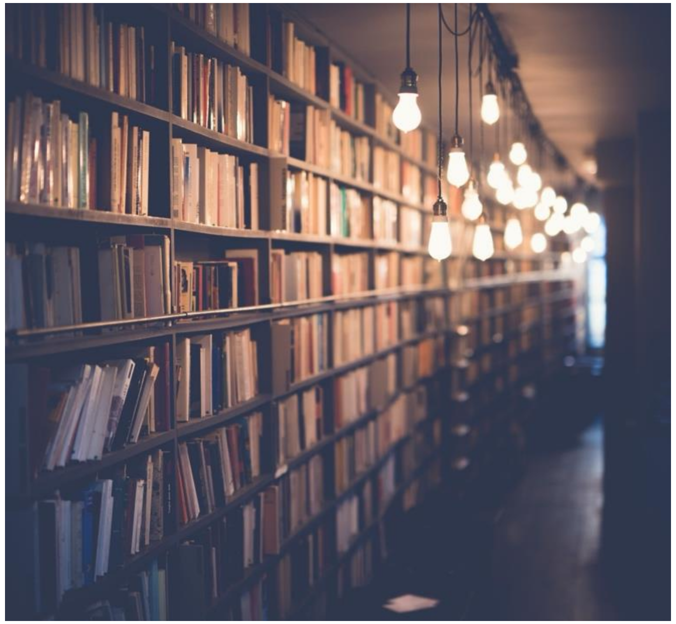
UiO : CEMO – Centre for Educational Measurement
University of Oslo

Meta-Analysis in Education

From Theory to Practice with R:
Publication Bias



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BEGIN.

Data Example: Gender Differences in Digital Literacy

(Campos et al., 2023; Scherer et al., 2024)

Meta-analytic data set of the gender differences in students' digital literacy

- Performance-based measures of digital literacy in K-12 student samples
- 59 effect sizes from 24 studies and 31 countries (both ILSA and non-ILSA)
- Standardized mean differences (female-male) Hedges' g (`g`) and sampling variance (`Var.g`)
- Identifiers of effect sizes (`ESID`), primary studies (`IDSTUDY`), and countries (`IDCOUNTRY`)
- Sample sizes (`N`, `nF`, `nM`)
- Contextual variables: Publication year (`PubYear`), availability of individual participant data (`IPD`), countries' power distance index (`PDI`), and GDP (`cGCP`)

Cross-Classified Data Structure

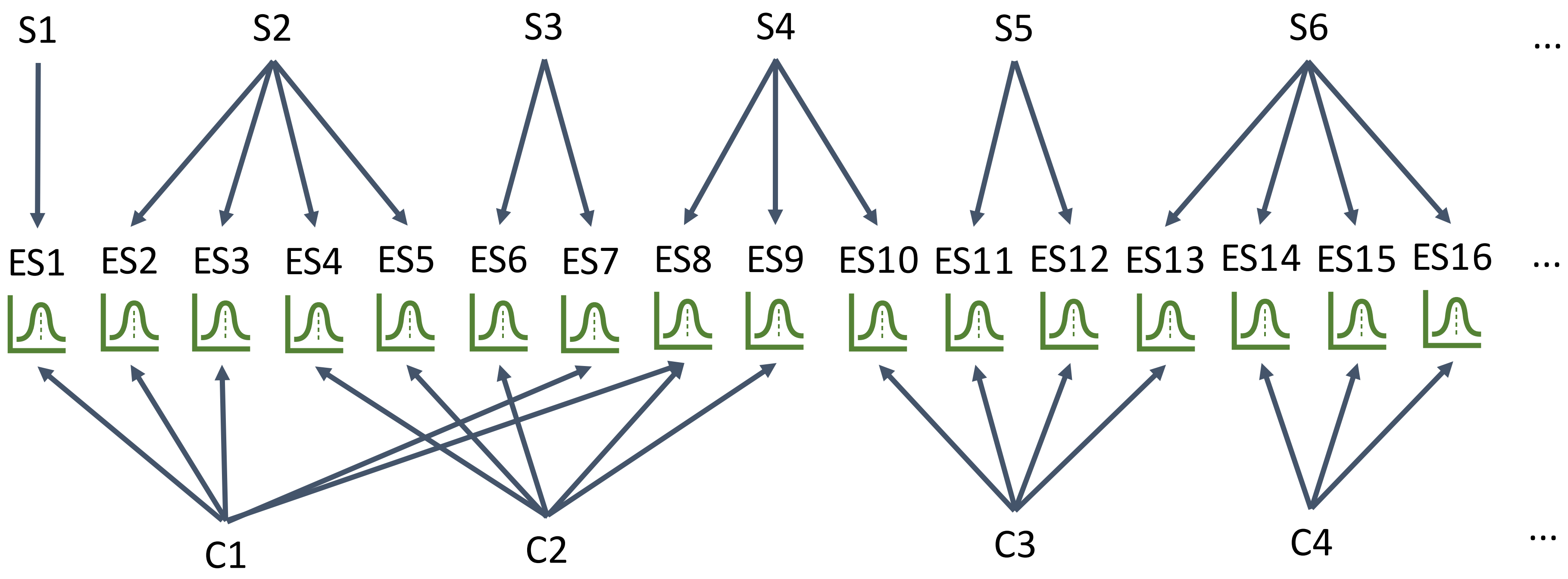
(CMM, 2019)

Four-level non-hierarchical structure with cross-classification

Studies (`StudyID`)
Between-study variance $\tau^2_{(3)}$
Within-study variance $\tau^2_{(2)}$

Effect sizes (`ESID`)
Sampling variance v_i

Countries (`IDCNTRY`)
Between-country variance $\tau^2_{(4)}$



Typical hierarchical structure if ILSA data are included.

Cross-Classified Random-Effects Model

Four-level random-effects model

Effect sizes nested in studies and countries

```r
## Model estimation
CCREM <- rma.mv(g,
                Var.g,
                random = list(~ 1 | IDSTUDY/ESID,
                              ~ 1 | IDCOUNTRY),
                method = "REML",
                data = dat,
                tdist = TRUE,
                test = "t",
                time = TRUE)
```

Independent nesting of effect sizes in studies and countries

(More details about CCREMs: Fernández-Castilla et al., 2019)

```
## Multivariate Meta-Analysis Model (k = 59; method: REML)
##
##    logLik  Deviance       AIC       BIC      AICc
##   33.8044  -67.6088  -59.6088  -51.3670  -58.8541
##
## Variance Components:
##
##              estim    sqrt  nlvls  fixed        factor
## sigma^2.1   0.0308  0.1754     24     no       IDSTUDY
## sigma^2.2   0.0016  0.0397     59     no  IDSTUDY/ESID
## sigma^2.3   0.0055  0.0743     31     no     IDCOUNTRY
##
## Test for Heterogeneity:
## Q(df = 58) = 592.4533, p-val < .0001
##
## Model Results:
##
## estimate      se    tval  df    pval   ci.lb   ci.ub
##   0.0951  0.0444  2.1424  58  0.0364  0.0062  0.1839  *
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$\tau^2_{(3)}$
$\tau^2_{(2)}$
$\tau^2_{(4)}$

β_R

Cross-Classified Random-Effects Model

Four-level random-effects model

Effect sizes nested in studies and countries

```
## Confidence intervals of the variance component(s)
confint.rma.mv(CCREM)
```

```
##              estimate  ci.lb  ci.ub
## sigma^2.1      0.0308 0.0112 0.0770
## sigma.1        0.1754 0.1057 0.2775
##
##              estimate  ci.lb  ci.ub
## sigma^2.2      0.0016 0.0000 0.0079
## sigma.2        0.0397 0.0000 0.0886
##
##              estimate  ci.lb  ci.ub
## sigma^2.3      0.0055 0.0003 0.0122
## sigma.3        0.0743 0.0175 0.1105
```

Level-specific I^2

```
## [1] 78.371363  4.008578 14.047175
```

$I^2_{(3)}$	$I^2_{(2)}$	$I^2_{(4)}$
Between studies	Within studies	Between countries

Publication Bias

Different tests address different aspects of publication bias

Publication Bias

Publication Bias is a Form of Non-Reporting Bias.

The probability of publishing a primary study is affected by its results: Statistically significant or hypothesis-confirming results are more likely to be published (see Harrer et al., 2021, chap. 9).

Detecting Publication Bias

- Funnel plot asymmetry and Egger's regression test
- Precision-effect test (PET) and precision-effect estimate with standard errors (PEESE)
- Funnel plot test
- Begg's correlation test
- Trim-and-fill analyses with the estimators L_0^+ and R_0^+
- Moderation by publication year and/or publication status (e.g., grey vs. published)
- Worst-case sensitivity analyses
- …

(e.g., Fernández-Castilla et al., 2021, https://doi.org/10.1080/00220973.2019.1582470;
Harrer et al., 2021, https://bookdown.org/MathiasHarrer/Doing_Meta_Analysis_in_R/pub-bias.html)

Detecting Publication Bias

Funnel Plot Symmetry

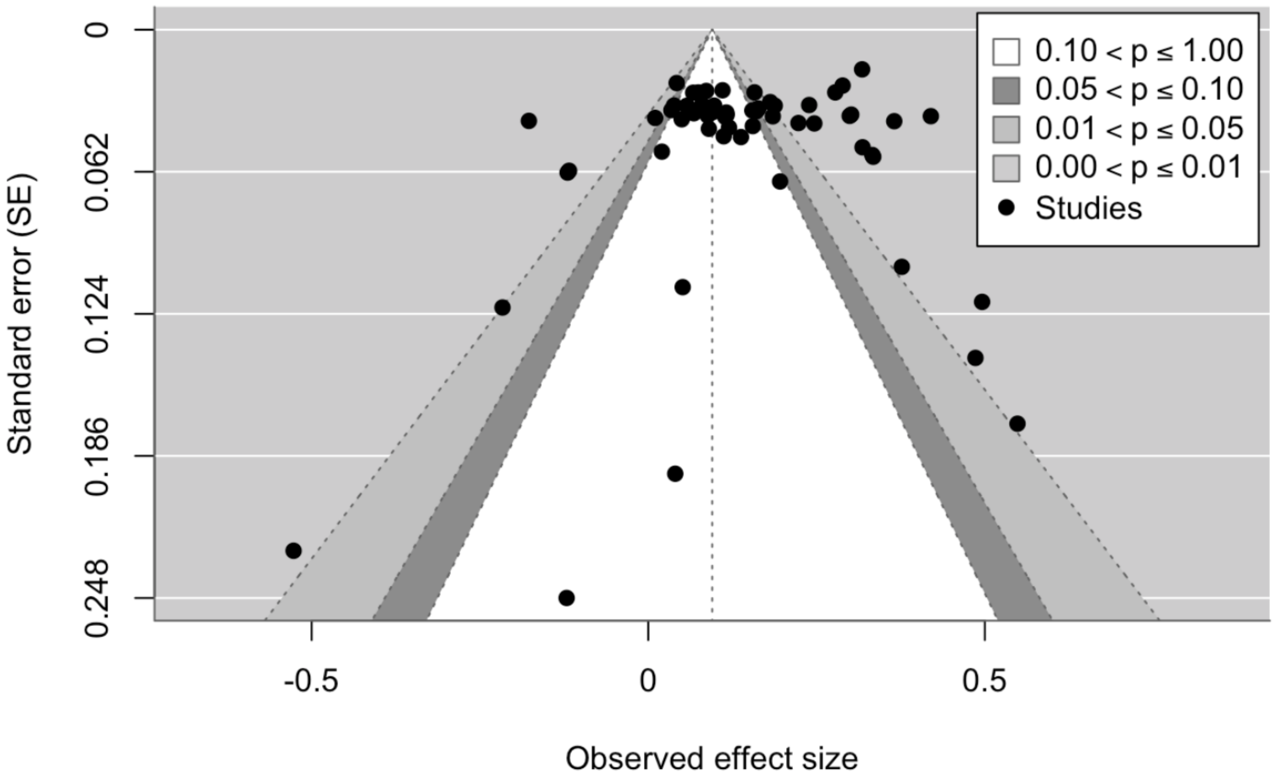
Plot the observed effect sizes against their standard errors to examine small-study effects

Graphical inspection of the plot asymmetry

Publication bias might be indicated by an asymmetric funnel plot.

```
metafor::funnel(CCREM,
                yaxis = "sei",
                level = c(90, 95, 99),
                shade = c("white", "gray55", "gray75"),
                legend = TRUE,
                xlab = "Observed effect size",
                ylab = "Standard error (SE)")
```

Contour-enhanced funnel plot



Detecting Publication Bias

Egger's Regression Test

Test the asymmetry of the funnel plot via a linear regression of the scaled effect sizes (z-score) on the precision ($1/SE$):

$$\frac{\hat{\theta}_i}{SE_{\hat{\theta}_i}} = \beta_0 + \beta_1 \frac{1}{SE_{\hat{\theta}_i}} + r_i$$

$$r_i \sim N(0, \sigma_r^2)$$

Test $\hat{\beta}_0$ against zero.

Significant $\hat{\beta}_0$ indicates funnel plot asymmetry.

Original Egger's test (Egger et al., 1997)

```
dat %>%
  mutate(y = g/sqrt(Var.g), x = 1/sqrt(Var.g)) %>%
  lm(y ~ x, data = .) %>%
  summary()
```

Modified Egger's test (Pustejovsky & Rodgers, 2019)

```
dat$SEg_modified <- sqrt((dat$nF + dat$nM)/(dat$nF*dat$nM))

dat %>%
  mutate(y = g/SEg_modified, x = 1/SEg_modified) %>%
  lm(y ~ x, data = .) %>%
  summary()
```

Detecting Publication Bias

Egger's Regression Test

Test the asymmetry of the funnel plot via a linear regression of the scaled effect sizes (z-score) on the precision ($1/SE$):

$$\frac{\hat{\theta}_i}{SE_{\hat{\theta}_i}} = \beta_0 + \beta_1 \frac{1}{SE_{\hat{\theta}_i}} + r_i$$

Test $\hat{\beta}_0$ against zero.

Significant $\hat{\beta}_0$ indicates funnel plot asymmetry.

Original Egger's test (Egger et al., 1997)

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.1390     1.0976  -1.038    0.304
## x             0.1945     0.0414   4.698  1.7e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.149 on 57 degrees of freedom
## Multiple R-squared:  0.2791, Adjusted R-squared:  0.2665
## F-statistic: 22.07 on 1 and 57 DF,  p-value: 1.701e-05
```

Result: No evidence of publication bias.

Detecting Publication Bias

Precision-effect test (PET) and Precision-effect estimate with standard error (PEESE)

Control for the sampling error (PET) or variance (PEESE) in the weighted average effect size and extract a limit effect (i.e., the effect with $SE = 0$).

Extract the intercepts from the two regressions:
PET estimate: $\hat{\theta}_{\mathrm{PET}} = \hat{\beta}_{0_{\mathrm{PET}}}$
PEESE estimate: $\hat{\theta}_{\mathrm{PEESE}} = \hat{\beta}_{0_{\mathrm{PEESE}}}$

Decision for an overall (controlled) estimate:

$$\hat{\theta}_{\mathrm{PEESE}} = \begin{cases} P(\hat{\beta}_{0_{\mathrm{PET}}} = 0) < 0.1 \text{ and } \hat{\beta}_{0_{\mathrm{PET}}} > 0: \; \hat{\beta}_{0_{\mathrm{PEESE}}} \\ \text{else: } \hat{\beta}_{0_{\mathrm{PET}}} \end{cases}$$

(Harrer et al., 2021, chap. 9; Stanley et al., 2014)

```
## Multilevel MEM
CCREM.pet <- rma.mv(g,
                    Var.g,
                    random = list(~ 1 | IDSTUDY/ESID,
                                  ~ 1 | IDCOUNTRY),
                    method = "REML",
                    data = dat,
                    tdist = TRUE,
                    test = "t",
                    mods =~ sqrt(Var.g))

# Summarize the results
summary(CCREM.pet, digits=4)
```

For PEESE: `mods =~ Var.g`

Detecting Publication Bias

Precision-effect test (PET) and Precision-effect estimate with standard error (PEESE)

```
## Multivariate Meta-Analysis Model (k = 59; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##  33.4927  -66.9854  -56.9854  -46.7702  -55.8090
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed        factor
## sigma^2.1  0.0330  0.1816     24     no       IDSTUDY
## sigma^2.2  0.0016  0.0401     59     no  IDSTUDY/ESID
## sigma^2.3  0.0055  0.0740     31     no     IDCOUNTRY
##
## Test for Residual Heterogeneity:
## QE(df = 57) = 583.2291, p-val < .0001
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 57) = 0.0872, p-val = 0.7688
##
## Model Results:
##
##              estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt        0.1111  0.0706   1.5732  57  0.1212  -0.0303  0.2525
## sqrt(Var.g)   -0.2225  0.7534  -0.2953  57  0.7688  -1.7312  1.2862
```

$\hat{\beta}_{0_{\mathrm{PET}}} = 0.11, p = .12$

Result: Decide for the PET estimate.

```
## Multivariate Meta-Analysis Model (k = 59; method: REML)
##
##   logLik  Deviance       AIC       BIC      AICc
##  33.8299  -67.6599  -57.6599  -47.4446  -56.4834
##
## Variance Components:
##
##             estim    sqrt  nlvls  fixed        factor
## sigma^2.1  0.0329  0.1814     24     no       IDSTUDY
## sigma^2.2  0.0016  0.0401     59     no  IDSTUDY/ESID
## sigma^2.3  0.0055  0.0739     31     no     IDCOUNTRY
##
## Test for Residual Heterogeneity:
## QE(df = 57) = 589.7801, p-val < .0001
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 57) = 0.6443, p-val = 0.4255
##
## Model Results:
##
##          estimate      se     tval  df    pval    ci.lb   ci.ub
## intrcpt    0.1167  0.0528   2.2119  57  0.0310   0.0110  0.2223  *
## Var.g     -2.6613  3.3155  -0.8027  57  0.4255  -9.3006  3.9779
```

$\hat{\beta}_{0_{\mathrm{PEESE}}} = 0.12, p = .03$

Detecting Publication Bias

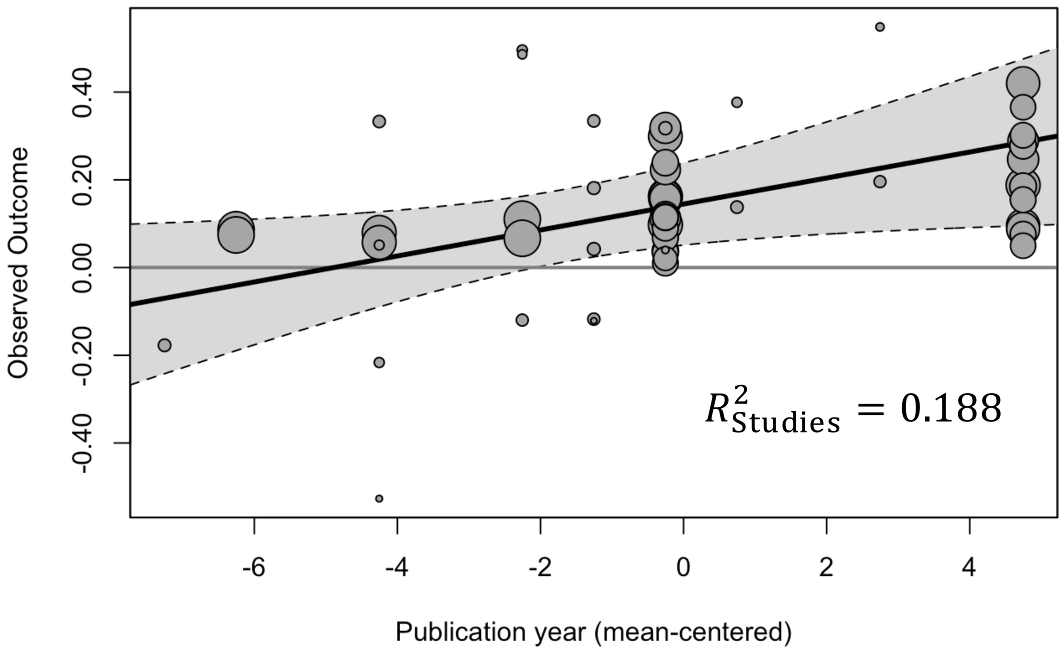
Moderation by publication characteristics

Examine the relation between publication characteristics (e.g., type of publication, peer-review status, publication year) and the effects.

Significant relations may indicate publication bias.

```
## Mixed-effects meta-regression model
CCREM.pub <- rma.mv(g,
                    Var.g,
                    random = list(~ 1 | IDSTUDY/ESID,
                                  ~ 1 | IDCOUNTRY),
                    method = "REML",
                    data = dat,
                    tdist = TRUE,
                    test = "t",
                    mods =~ scale(PubYear, ## mean-centered
                                  center = TRUE,
                                  scale = FALSE))
```

$\hat{\beta}_1 = 0.03, SE = 0.01, 95\ \%CI\ [0.01, 0.06]$



Result: Significant relation between publication year and the effects. More recent studies exhibit larger effect sizes.

Publication Bias

Analysis of Publication/Selection Bias in Multilevel Meta-Analysis

Several methods that can be based on a working model:

- Egger's regression test ("precision-effect test"; PET)
- Precision-effect estimate with standard errors (PEESE)
- Funnel plot test
- Begg's correlation test
- Trim-and-fill analyses with the estimators L_0^+ and R_0^+
- Moderation by publication year and/or publication status (e.g., grey vs. published)

(see Fernández-Castilla et al., 2021, https://doi.org/10.1080/00220973.2019.1582470;
Harrer et al., 2021, https://bookdown.org/MathiasHarrer/Doing_Meta_Analysis_in_R/pub-bias.html)


The End.

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